تمارين شاملة و تقييمية للحصص السابقة

لكل من الدوال التالية: احسب الدالة المشتقة $f'(x)$

1. $f(x) = 2x^2 - 16x + 5$
2. $k(x) = 363.5x^2 + x - 10$
3. $g(x) = -2x^3 - x + 9$
4. $h(x) = 71.5x^2 - 22x + 100$
5. $U(x) = -91.5x^2 - 22x + 3$
6. $V(x) = 272x^2 - 21x + 4$
7. $f(x) = \sqrt{x}$
8. $f(x) = \sqrt{2x + 6}$
9. $f(x) = \sqrt{-3x + 12}$
10. $f(x) = \sqrt{x^2 + 4}$
11. $f(x) = \sqrt{x^2 - 6x + 10}$
12. $f(x) = \sqrt{3x^2 + 2x + 1}$
13. $f(x) = \sqrt{-x^2 + 9}$
14. $f(x) = x\sqrt{x}$
15. $f(x) = \frac{\sqrt{x}}{x+1}$
16. $f(x) = 10x - 45$
17. $f(x) = -15x + 200$
18. $f(x) = x^4 - 2x^2 + 3$
19. $f(x) = -x^4 + 8x^2 + 1$
20. $f(x) = \frac{x+2}{x-1}$
21. $f(x) = \frac{2x-3}{x+4}$
22. $f(x) = \frac{-x+5}{3x+1}$
23. $f(x) = 3x + 2 + \frac{1}{x-2}$
24. $f(x) = -x + 1 - \frac{2}{x+3}$
25. $f(x) = \sqrt{5x - 1}$
26. $f(x) = \sqrt{-4x^2 + 16}$
27. $f(x) = \sqrt{x^3 + 1}$
28. $f(x) = \sqrt{2x^2 - 8x + 11}$
29. $f(x) = (x - 2)\sqrt{x}$
30. $f(x) = \sqrt{\frac{x}{x+1}}$
31. $f(x) = -5x^2 - 10x + 3$
32. $f(x) = 2x^3 - 9x^2 + 12x - 4$
33. $f(x) = -x^3 + 27x - 10$
34. $f(x) = \frac{3x-1}{2x+5}$
35. $f(x) = \frac{4-x}{x+1}$
36. $f(x) = 5x - 7 + \frac{3}{x-1}$
37. $f(x) = -2x + 4 + \frac{1}{x+2}$
38. $f(x) = \sqrt{x^2 + 16}$
39. $f(x) = \sqrt{4x - 8}$
40. $f(x) = \sqrt{-2x^2 + 8x + 10}$

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41. $f(x) = \sqrt{x^2 - 3x}$

42. $f(x) = \frac{1}{\sqrt{x}}$

43. $f(x) = \frac{1}{\sqrt{2x+3}}$

44. $f(x) = x + \sqrt{x}$

45. $f(x) = 2x - \sqrt{x-1}$

46. $f(x) = \frac{-3x+2}{x-5}$

47. $f(x) = 4x + 1 - \frac{5}{x+4}$

48. $f(x) = -3x + 9 - \frac{2}{x-6}$

49. $f(x) = \sqrt{x^4 + 1}$

50. $f(x) = \sqrt{-x^3 + 12x}$

الحلول

المشتقات

1. $f'(x) = 4x - 16$
2. $k'(x) = 727x + 1$
3. $g'(x) = -6x^2 - 1$
4. $h'(x) = 143x - 22$
5. $U'(x) = -183x - 22$
6. $V'(x) = 544x - 21$
7. $f'(x) = \frac{1}{2\sqrt{x}}$
8. $f'(x) = \frac{2}{2\sqrt{2x+6}} = \frac{1}{\sqrt{2x+6}}$
9. $f'(x) = \frac{-3}{2\sqrt{-3x+12}}$
10. $f'(x) = \frac{2x}{2\sqrt{x^2+4}} = \frac{x}{\sqrt{x^2+4}}$
11. $f'(x) = \frac{2x-6}{2\sqrt{x^2-6x+10}} = \frac{x-3}{\sqrt{x^2-6x+10}}$
12. $f'(x) = \frac{6x+2}{2\sqrt{3x^2+2x+1}} = \frac{3x+1}{\sqrt{3x^2+2x+1}}$
13. $f'(x) = \frac{-2x}{2\sqrt{-x^2+9}} = \frac{-x}{\sqrt{-x^2+9}}$
14. $f'(x) = \sqrt{x} + \frac{x}{2\sqrt{x}} = \frac{3}{2}\sqrt{x}$
15. $f'(x) = \frac{\frac{1}{2\sqrt{x}}(x+1) - \sqrt{x}}{(x+1)^2} = \frac{1-x}{2\sqrt{x}(x+1)^2}$
16. $f'(x) = 10$
17. $f'(x) = -15$

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18. $f'(x) = 4x^3 - 4x$

19. $f'(x) = -4x^3 + 16x$

20. $f'(x) = \frac{-3}{(x-1)^2}$

21. $f'(x) = \frac{11}{(x+4)^2}$

22. $f'(x) = \frac{-16}{(3x+1)^2}$

23. $f'(x) = 3 - \frac{1}{(x-2)^2}$

24. $f'(x) = -1 + \frac{2}{(x+3)^2}$

25. $f'(x) = \frac{5}{2\sqrt{5x-1}}$

26. $f'(x) = \frac{-8x}{2\sqrt{-4x^2+16}} = \frac{-4x}{\sqrt{-4x^2+16}}$

27. $f'(x) = \frac{3x^2}{2\sqrt{x^3+1}}$

28. $f'(x) = \frac{4x-8}{2\sqrt{2x^2-8x+11}} = \frac{2x-4}{\sqrt{2x^2-8x+11}}$

29. $f'(x) = \sqrt{x} + \frac{x-2}{2\sqrt{x}} = \frac{3x-2}{2\sqrt{x}}$

30. $f'(x) = \frac{1}{2\sqrt{\frac{x}{x+1}}} \cdot \frac{1}{(x+1)^2}$

31. $f'(x) = -10x - 10$

32. $f'(x) = 6x^2 - 18x + 12$

33. $f'(x) = -3x^2 + 27$

34. $f'(x) = \frac{17}{(2x+5)^2}$

35. $f'(x) = \frac{-5}{(x+1)^2}$

36. $f'(x) = 5 - \frac{3}{(x-1)^2}$

37. $f'(x) = -2 - \frac{1}{(x+2)^2}$

38. $f'(x) = \frac{x}{\sqrt{x^2+16}}$

39. $f'(x) = \frac{4}{2\sqrt{4x-8}} = \frac{2}{\sqrt{4x-8}}$

40. $f'(x) = \frac{-4x+8}{2\sqrt{-2x^2+8x+10}} = \frac{-2x+4}{\sqrt{-2x^2+8x+10}}$

41. $f'(x) = \frac{2x-3}{2\sqrt{x^2-3x}}$

42. $f'(x) = -\frac{1}{2x\sqrt{x}}$

43. $f'(x) = \frac{-2}{2(2x+3)\sqrt{2x+3}} = -\frac{1}{(2x+3)\sqrt{2x+3}}$

44. $f'(x) = 1 + \frac{1}{2\sqrt{x}}$

45. $f'(x) = 2 - \frac{1}{2\sqrt{x-1}}$

46. $f'(x) = \frac{13}{(x-5)^2}$

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47. $f'(x) = 4 + \frac{5}{(x+4)^2}$

48. $f'(x) = -3 + \frac{2}{(x-6)^2}$

49. $f'(x) = \frac{4x^3}{2\sqrt{x^4+1}} = \frac{2x^3}{\sqrt{x^4+1}}$

$$f'(x) = \frac{-3x^2 + 12}{2\sqrt{-x^3 + 12x}}$$